



**EAST WEST UNIVERSITY**

**Project Title: Digital Farming Assistant**

**Group: Alpha Matrix**

**Submitted To**

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**Course Details**

**Course Title:** IT Project Management and Entrepreneurship

**Course code:** CSE495

**Section:** 04

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## Table of Contents

| <b>Serial</b> | <b>Contents</b>      | <b>Pages</b> |
|---------------|----------------------|--------------|
| 1             | User Story           | 3-6          |
| 2             | Stakeholder Register | 7-10         |
| 3             | Project Charter      | 11           |
| 4             | WBS                  | 12-14        |
| 5             | Product Backlog      | 15           |

## User Story

A dedicated Farmer, logs into the Digital Farming Assistant using his Account ID. His Name, Phone Number, and Language Preference were set during Registration Date, and his Authentication Method is simple, just simple password, and his phone number. His Profile Image appears on the dashboard, and his Offline Access Enabled flag ensures he can use the app even in low-connectivity zones. His Location ID is synced to provide localized insights, and his Crop history, Diagnosis history, Forum Posts, and Saved Crop Profiles are stored under his Farmer ID.

In the background, the Admin team, identified by Admin ID manages the system. Each admin has a Name, Email, and Phone Number, and their Role (e.g., Moderator or SuperAdmin) defines their access level. They monitor System Update Logs, track Authentication Logs, handle Forum Moderation Actions, and manage User Management Actions. Their Last Login timestamp ensures accountability.

Rafiq heads to his field and captures an Image of his crops using his smartphone. The image is tagged with an Image ID, linked to his FarmerID, and includes a Timestamp, File Path, and Crop Type. The system detects Symptoms Detected and stores the Field Location. Since the farmer is offline, the Upload Status is marked as “queued” and the Offline Flag is set to true.

The Image Analysis Service, part of the AI Assistant, processes the image. The assistant, identified by Assistant ID, uses Tool Used (e.g., ML model) to generate a ResponseContent with a ConfidenceScore. The Service Type is “ImageAnalysis,” and the Request ID links the session. The result is tied to a Linked Diagnosis ID.

A new Diagnosis is created with a unique DiagnosisID, linked to the Image ID, Farmer ID, and Crop ID. It detects a Disease - say, a fungal rot and records the Disease Detected, Severity Level, Accuracy Score, and Description. The Timestamp logs the moment of analysis, and the system provides a Recommendation. An Agronomist, identified by Agronomist ID, later verifies the result. Their Name, Phone Number, Expertise Area (e.g., pest control), and Verified Status are stored. They add Recommendations Given and Forum Comments to the system. Their Profile Image, Location, and Language Preference help tailor their interface.

The Disease entity includes its Name, Symptoms, SeverityScale, AffectedCrops, and TreatmentOptions. The system flags a Pest Outbreak Flag, estimates the Rot Percentage Range, and logs the Detection Accuracy and Prevention Tips.

Meanwhile, the Weather API feeds Weather Data into the app. Each record has a Weather Data ID, Location, Date, Temperature, Humidity, Rainfall, Wind Speed, and Forecast Type (daily/weekly). The Source is OpenWeatherMap, and each entry has a Timestamp. If a storm is predicted, a Weather Alert is triggered with an AlertType, and a Notification is sent. The Notification ID, Recipient ID, Type (e.g., WeatherAlert), Message Content, Timestamp, ReadStatus, and Delivery Method (App, Email, SMS) are all logged.

The AI Assistant also provides Guidance on Cultivation. Rafiq's Cultivation ID links to his Farmer ID and Crop ID, with attributes like Start Date, End Date, Field Location, Irrigation Schedule, Harvesting Date, Soil Type, and Yield Estimate. The assistant helps him Schedule irrigation based on soil moisture and forecast data.

Curious about new crops, Rafiq explores the Crop Library. Each Crop has a Crop ID, Name, Season, Soil Type, Growth Duration, Common Diseases, Recommended Fertilizers, Recommended Pesticides, Tips, and an Image Reference. He filters using Keyword Tags and views Crop Profiles with Profile ID, Planting Method, Soil Requirement, Fertilizer Recommendation, Pesticide Recommendation, Field Layout Guide, and Offline Saved Flag. The Last Viewed Date and Created By (Farmer or Agronomist) are tracked.

In the evening, Rafiq joins the Community Forum. He creates a Post ID with an Image Attached, writes Post Content, and adds a Timestamp. Other farmers and agronomists leave Comments, click Likes, and reply. Each post is tagged with KeywordTags, marked with a Searchable Flag, and linked to a Forum ID.

Behind the scenes, the Authentication system logs each Auth ID, User ID, Phone Number, OTP Status, Last Login Timestamp, Device Info, Language Preference, Forgot Password Flag, and Auth Method.

By the end of the day, Rafiq feels supported. His Account — with its Account ID, User Type, Name, Phone Number, Language Preference, Registration Date, Profile Image, Last Login, Offline Access Enabled, and Authentication Method — connects him to a system that understands his needs.

| Features and sub features                                                                                                                                                                                                                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Login System</p> <p>Sub-features includes :</p> <ul style="list-style-type: none"> <li>i) Name</li> <li>ii) phone number</li> <li>iii) preferred language</li> </ul>                                                                                  | <p>This is our first and small features and Simple authentication using name and phone number only (no email and password, no otp complexity). Also, they can choose the language they want to use.</p>                                                                                                                                                                                                                                                                                                   |
| <p>Crop health Checker</p> <p>Sub-features include:</p> <ul style="list-style-type: none"> <li>i) Image Capture &amp; Upload</li> <li>ii) Diagnosis Summary</li> <li>iii) Offline Image Queue</li> <li>iv) Diagnosis History</li> </ul>                  | <p>Uses trained machine learning models to analyze crop images and detect the percentage of rot, pest attack, or disease, then provides recommendations for treatment or care. They can click the pictures before and upload it in the offline image queue because of internet connectivity and also check previous images and feedbacks in history.</p>                                                                                                                                                  |
| <p>Weather Forecast</p> <p>Sub-features includes :</p> <ul style="list-style-type: none"> <li>i) daily and weekly forecast view</li> <li>ii) Location-Based Forecast</li> <li>iii) Farm Planning advisory</li> </ul>                                     | <p>Displays real-time and weekly weather forecasts using integrated APIs(Open Weather Map) which will help farmers decide irrigation or harvesting schedules. Also, weather alerts (heavy rainfall,storm) will be there through notifications. Location-based forecast details will also be there to give accurate insights to farmers and Farm Planning advisory based on the upcoming weather suggest which crop will be better to grow now or should crop be given water, irrigation advices etc .</p> |
| <p>Crop Library</p> <p>Sub-feature includes :</p> <ul style="list-style-type: none"> <li>i) Filter-Based Search</li> <li>ii) Crop Profiles (<b>Planting methods,Soil requirements</b>,recommend fertilizers etc)</li> <li>iii) Offline Access</li> </ul> | <p>A searchable database of crops with planting methods, soil requirements, fertilizers, and pest control guidelines., filter-based search by seasons and soils, moreover saving option for most viewed crops which enables offline accessibility.</p>                                                                                                                                                                                                                                                    |
| <p>Community Forum</p> <p>Sub-feature includes :</p> <ul style="list-style-type: none"> <li>i) post with image</li> <li>ii) comment</li> <li>iii) like</li> <li>iv) search questions by keyword</li> </ul>                                               | <p>A discussion board where farmers can ask questions, share experiences, and get expert suggestions. Here farmers ca post anything with image, comment section is also here where they can receive guidance by the fellow farmers, like system is here to show the relevant feedback and search by keyword option helps to find existing problem and it's solutions.</p>                                                                                                                                 |

## **Assumptions and Constraints**

### **Assumptions:**

- Most users have smartphones or at least basic internet access.
- Users can easily access the application and its online features.
- Reliable weather API services are available for accurate and timely updates.
- Farmers are able to capture clear images of their crops using mobile phones.
- The AI system can effectively detect crop diseases from these images.
- The initial AI model is trained on a diverse, regionally relevant dataset.
- The system can recognize common crops and conditions across different areas.

### **Constraints:**

- AI accuracy may be lower for some crops due to limited training data.
- The application depends on internet connectivity for:
  - Real-time weather updates
  - Access to the community forum
- Poor network coverage in rural areas may limit usability.
- Limited technical literacy among farmers may slow adoption.
- Managing large volumes of uploaded crop images is challenging.
- Cloud storage integration is required for efficient data handling.

## Stakeholder Register – Digital Farming Assistant

### 1. Stakeholder Identification Information

| ID | Name/Role                              | Organization                         | Contact        | Position                                 |
|----|----------------------------------------|--------------------------------------|----------------|------------------------------------------|
| S1 | Project Sponsor                        | East West University                 | sponsor@ewu.bd | Final academic authority, approves scope |
| S2 | Project Manager                        | Project Team                         | pm@dfa.com     | Leads execution and coordination         |
| S3 | HR Consultant (Dr. Rajesh Chakraborty) | Shere Bangla Agricultural University |                | Advisory                                 |
| S4 | Development Team                       | EWU Students                         |                | Designs and develops system              |
| S5 | Farmer Representative                  | Farming Community                    |                | Represents end users                     |
| S6 | Agricultural Expert                    | Agriculture Sector                   | agri@expert.bd | Provides crop guidance knowledge         |
| S7 | Weather Data Provider                  | Meteorological Dept                  | weather@met.bd | Supplies forecast data                   |
| S8 | Government Agriculture Officer         | Govt. of Bangladesh                  | govt@agri.bd   | govt@agri.bd                             |

## 2. Assessment Information

| <b>ID</b> | <b>Main Requirements</b>   | <b>Expectations</b>            | <b>Influence Level</b> | <b>Interest Level</b> | <b>Key Phase Interest</b> |
|-----------|----------------------------|--------------------------------|------------------------|-----------------------|---------------------------|
| S1        | On-time completion         | Successful academic innovation | Very High              | Medium                | Initiation, Closing       |
| S2        | Clear scope & control      | Project success                | High                   | Very High             | All phases                |
| S3        | Proper documentation       | Academic quality output        | Medium                 | Medium                | Planning                  |
| S4        | Clear requirements         | Smooth development             | High                   | High                  | Execution                 |
| S5        | Simple mobile interface    | Better crop yield & alerts     | Medium                 | Very High             | Execution                 |
| S6        | Accurate recommendations   | Reliable farming guidance      | Medium                 | High                  | Planning, Execution       |
| S7        | Reliable API data          | Proper integration             | High                   | Medium                | Execution                 |
| S8        | Data protection compliance | Secure farmer data             | Medium                 | Medium                | Monitoring                |



### 3. Stakeholder Classification

| <b>ID</b> | <b>Type</b> | <b>Attitude</b> | <b>Category</b>          |
|-----------|-------------|-----------------|--------------------------|
| S1        | Internal    | Positive        | Key Sponsor              |
| S2        | Internal    | Positive        | Core Team                |
| S3        | Internal    | Neutral         | Academic Advisor         |
| S4        | Internal    | Positive        | Development Team         |
| S5        | External    | Positive        | Primary User Stakeholder |
| S6        | External    | Supportive      | Subject Expert           |
| S7        | External    | Neutral         | Data Provider            |
| S8        | External    | Cautious        | Regulatory Authority     |

#### 4. Stakeholder Management Strategy

| ID | Strategy                              | Communication Plan              | Risk Handling                    |
|----|---------------------------------------|---------------------------------|----------------------------------|
| S1 | Keep satisfied, involve in milestones | Monthly progress report         | Early warning on delays          |
| S2 | High engagement and authority         | Weekly team meetings            | Escalate conflicts quickly       |
| S3 | Consult during documentation review   | Milestone-based review meetings | Validate academic requirements   |
| S4 | Active coordination                   | Daily/weekly development sync   | Monitor backlog                  |
| S5 | Engage through testing & feedback     | User demo sessions              | Simplify UI, training support    |
| S6 | Involve in crop logic validation      | Requirement workshops           | Verify recommendation accuracy   |
| S7 | Formal API agreement                  | Integration progress updates    | Backup data source               |
| S8 | Ensure compliance with policies       | Periodic compliance review      | Implement data security measures |

## Project Charter

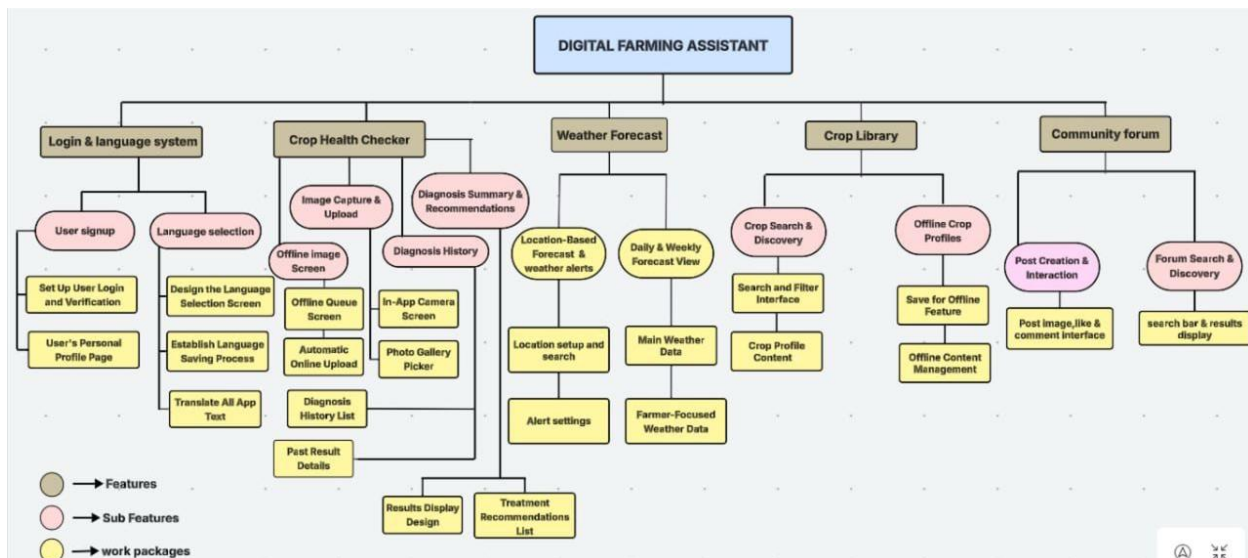
| Project Charter                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Project Title                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Digital Farming Assistant |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |
| Start Date                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | —                         | Project Manager                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | —                    |
| End Date                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | —                         | Sponsor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | East West University |
| Project Description And Need                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |
| The Digital Farming Assistant aims to enhance agricultural efficiency and productivity. It will help increase crop yield by providing timely weather forecasts to alert farmers about upcoming natural disasters, and by offering guidance to protect crops from various threats. In addition, the system will include a community forum where farmers can share their problems, experiences, and creative ideas with others and knowledge exchange within the farming community. |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |
| Project Scope                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                           | Risk And Issues                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |
| 1. Develop a Digital Farming Assistant system for Bangladeshi farmers.<br>2. Provide real-time weather forecasts and disaster alerts.<br>3. Recommend suitable crops and fertilizers based on region and season.<br>4. Create a community forum for farmers to share ideas and problems.<br>5. Implement a secure login system for user access and data protection.                                                                                                               |                           | 1. Unauthorized access or data breach due to weak security in the login system.<br>2. Inaccurate or outdated agricultural and weather data.<br>3. Low user adoption due to limited digital literacy among farmers.<br>4. Poor internet connectivity in rural areas.                                                                                                                                                                                                                    |                      |
| Deliverables                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                           | Assumptions                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                      |
| 1. Prepare the system design and development plan for the Digital Farming Assistant.<br>2. Development of the application with all core modules (weather, crop recommendation, community forum, and login system).<br>3. Testing and debugging of the system to ensure functionality and reliability.<br>4. Preparation of user manual and project documentation.                                                                                                                 |                           | 1. All farmers will have access to smartphones and internet connectivity.<br>2. All users will actively use the system for agricultural guidance and information sharing.<br>3. All weather and agricultural data sources will remain accurate and regularly updated.<br>4. All farmers will understand and use the system primarily in Bangla language.<br>5. All system modules (forecasting, crop recommendation, and forum) will function as intended during the project timeline. |                      |
| Budget                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |
| No monetary value included, it is only for educational purpose.                                                                                                                                                                                                                                                                                                                                                                                                                   |                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                      |
| Milestone                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Target Date of completion |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Actual Date          |
| Project prototype                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | —                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | —                    |
| HR Consultants                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Dr. Rajesh Chakraborty    | Sponsor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | East West University |
| Team Members                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | —                         | East West University                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                      |

## Works Breakdown Structure Diagram (WBS)

A Work Breakdown Structure (WBS) diagram is a visual representation that organizes a project into smaller, more manageable components. It takes the overall scope of work and breaks it down into hierarchical levels, starting from the main deliverable and moving into sub-deliverables, tasks, and work packages. By showing this structure in a diagram, it becomes easier to see how different parts of the project connect, how responsibilities can be assigned, and how progress can be tracked.

The purpose of a WBS diagram is to provide clarity and control over complex projects. It helps everyone involved understand exactly what needs to be done, how the work is divided, and where their role fits within the bigger picture. It reduces ambiguity by defining tasks clearly, supports better estimation of time and resources, and ensures that nothing important is overlooked. For stakeholders, it acts as a communication tool that aligns expectations, while for project managers, it serves as a roadmap to plan, monitor, and deliver the project successfully. In essence, the WBS diagram transforms a large, overwhelming project into a structured plan that is easier to execute and manage.

### WBS Diagram :



## **1. Login & Language System**

- **1.1 User Signup**
  - **1.1.1** Set Up User Login and Verification
  - **1.1.2** User's Personal Profile Page
- **1.2 Language Selection**
  - **1.2.1** Design the Language Selection Screen
  - **1.2.2** Establish Language Saving Process
  - **1.2.3** Translate All App Text

## **2. Crop Health Checker**

- **2.1 Image Capture & Upload**
  - **2.1.1** Offline Image Screen
  - **2.1.2** Offline Queue Screen
  - **2.1.3** Automatic Online Upload
  - **2.1.4** In-App Camera Screen
  - **2.1.5** Photo Gallery Picker
- **2.2 Diagnosis Summary & Recommendations**
  - **2.2.1** Diagnosis History
  - **2.2.2** Diagnosis History List
  - **2.2.3** Past Result Details
  - **2.2.4** Results Display Design
  - **2.2.5** Treatment Recommendations List

## **3. Weather Forecast**

- **3.1 Location-Based Forecast & Weather Alerts**
  - **3.1.1** Location Setup and Search
  - **3.1.2** Alert Settings
- **3.2 Daily & Weekly Forecast View**
  - **3.2.1** Main Weather Data
  - **3.2.2** Farmer-Focused Weather Data

## **4. Crop Library**

- **4.1 Crop Search & Discovery**
  - 4.1.1 Search and Filter Interface
  - 4.1.2 Crop Profile Content
- **4.2 Offline Crop Profiles**
  - 4.2.1 Save for Offline Feature
  - 4.2.2 Offline Content Management

## **5. Community Forum**

- **5.1 Post Creation & Interaction**
  - 5.1.1 Post Image, Like & Comment Interface
- **5.2 Forum Search & Discovery**
  - 5.2.1 Search Bar & Results Display

# Product Backlog

|    | A                   | B   | C                                              | D       | E       | F     | G       | H      | I       | J      | K       | L      | M       | N      |
|----|---------------------|-----|------------------------------------------------|---------|---------|-------|---------|--------|---------|--------|---------|--------|---------|--------|
|    | Module              | ID  | User Stories                                   | Initial | Sprint# | Total | S1 done | S1 rem | S2 done | S2 rem | S3 done | S3 rem | S4 done | S4 rem |
| 1  | User Management     | 1.1 | user, create account using name & phone number | 14      | 1       | 60    | 14      | 0      | 0       | 0      | 0       | 0      | 0       | 0      |
| 2  | User Management     | 1.2 | user, login using phone number                 | 8       | 1       | 60    | 8       | 0      | 0       | 0      | 0       | 0      | 0       | 0      |
| 3  | User Management     | 1.3 | user, select preferred language                | 6       | 1       | 60    | 6       | 0      | 0       | 0      | 0       | 0      | 0       | 0      |
| 4  | User Management     | 1.4 | user, view profile & activity history          | 12      | 1       | 60    | 12      | 0      | 0       | 0      | 0       | 0      | 0       | 0      |
| 5  | User Management     | 1.5 | user, enable offline access                    | 10      | 1       | 60    | 10      | 0      | 0       | 0      | 0       | 0      | 0       | 0      |
| 6  |                     |     |                                                |         |         |       |         |        |         |        |         |        |         |        |
| 7  |                     |     |                                                |         |         |       |         |        |         |        |         |        |         |        |
| 8  | Crop Health Checker | 2.1 | user, capture crop image                       | 12      | 2       | 85    | 0       | 12     | 8       | 4      | 4       | 0      | 0       | 0      |
| 9  | Crop Health Checker | 2.2 | user, upload image (online/offline)            | 16      | 2       | 85    | 0       | 16     | 10      | 6      | 6       | 0      | 0       | 0      |
| 10 | Crop Health Checker | 2.3 | system, offline image queue handling           | 18      | 2       | 85    | 0       | 18     | 12      | 6      | 6       | 0      | 0       | 0      |
| 11 | Crop Health Checker | 2.4 | system, AI-based image analysis                | 30      | 3       | 95    | 0       | 30     | 0       | 30     | 15      | 15     | 15      | 0      |
| 12 | Crop Health Checker | 2.5 | system, show diagnosis & recommendations       | 18      | 3       | 95    | 0       | 18     | 0       | 18     | 10      | 8      | 8       | 0      |
| 13 | Crop Health Checker | 2.6 | user, view diagnosis history                   | 10      | 3       | 95    | 0       | 10     | 0       | 10     | 6       | 4      | 4       | 0      |
| 14 |                     |     |                                                |         |         |       |         |        |         |        |         |        |         |        |
| 15 | Weather Forecast    | 3.1 | user, location-based weather data              | 14      | 2       | 70    | 0       | 14     | 9       | 5      | 5       | 0      | 0       | 0      |
| 16 | Weather Forecast    | 3.2 | user, daily & weekly forecast view             | 12      | 2       | 70    | 0       | 12     | 8       | 4      | 4       | 0      | 0       | 0      |
| 17 | Weather Forecast    | 3.3 | system, weather alerts generation              | 10      | 3       | 70    | 0       | 10     | 0       | 10     | 6       | 4      | 4       | 0      |
| 18 | Weather Forecast    | 3.4 | system, send notifications                     | 10      | 3       | 70    | 0       | 10     | 0       | 10     | 6       | 4      | 4       | 0      |
| 19 | Weather Forecast    | 3.5 | system, farming advice based on weather        | 16      | 4       | 70    | 0       | 16     | 0       | 16     | 0       | 16     | 9       | 7      |
| 20 |                     |     |                                                |         |         |       |         |        |         |        |         |        |         |        |
| 21 | Crop Library        | 4.1 | user, search crops by filter                   | 14      | 3       | 65    | 0       | 14     | 0       | 14     | 8       | 6      | 6       | 0      |
| 22 | Crop Library        | 4.2 | user, view crop profiles                       | 18      | 3       | 65    | 0       | 18     | 0       | 18     | 10      | 8      | 8       | 0      |
| 23 | Crop Library        | 4.3 | user, save crops offline                       | 10      | 4       | 65    | 0       | 10     | 0       | 10     | 0       | 10     | 6       | 4      |
| 24 | Crop Library        | 4.4 | system, manage crop data                       | 14      | 4       | 65    | 0       | 14     | 0       | 14     | 0       | 14     | 8       | 6      |
| 25 |                     |     |                                                |         |         |       |         |        |         |        |         |        |         |        |
| 26 | Community Forum     | 5.1 | user, create post with image                   | 16      | 4       | 60    | 0       | 16     | 0       | 16     | 0       | 16     | 9       | 7      |
| 27 | Community Forum     | 5.2 | user, comment on posts                         | 10      | 4       | 60    | 0       | 10     | 0       | 10     | 0       | 10     | 6       | 4      |
| 28 | Community Forum     | 5.3 | user, like posts                               | 6       | 4       | 60    | 0       | 6      | 0       | 6      | 0       | 6      | 4       | 2      |
| 29 | Community Forum     | 5.4 | user, search posts by keyword                  | 12      | 4       | 60    | 0       | 12     | 0       | 12     | 0       | 12     | 7       | 5      |
| 30 |                     |     |                                                |         |         |       |         |        |         |        |         |        |         |        |
| 31 | Admin System        | 6.1 | admin, manage users                            | 14      | 4       | 60    | 0       | 14     | 0       | 14     | 0       | 14     | 8       | 6      |
| 32 | Admin System        | 6.2 | admin, monitor logs & activities               | 10      | 4       | 60    | 0       | 10     | 0       | 10     | 0       | 10     | 6       | 4      |
| 33 | Admin System        | 6.3 | system, maintain authentication logs           | 10      | 4       | 60    | 0       | 10     | 0       | 10     | 0       | 10     | 5       | 5      |
| 34 |                     |     |                                                |         |         |       |         |        |         |        |         |        |         |        |
| 35 |                     |     |                                                | 350     |         |       | 50      |        | 47      |        | 86      |        | 117     |        |